

Preparing for ISEF Interviews

In order for you to come to a decision as to how you will rank the different projects that you will see at ISEF, the most important factor is the interview, where you will spend 10-12 minutes speaking with the student or students who produced the project. However, the key to making those few minutes as productive as possible is your preparation for the interviews.

Resources

In “the old days,” ISEF judges had essentially no information about the projects they were going to evaluate in advance of the fair. At most, judges could read the project abstract, but these were unformatted introductions that frequently did not even describe what the student had done. However, now ISEF judges have the advantage of a wealth of helpful insight into the projects before meeting the students for the all-important interview. These resources are summarized here and are described in more detail in Document B: *Materials Provided by the Students* within Section II of the *Judge Information Board*.

Online Resources

These resources will be available on ProjectBoard, posted for your inspection ten days to two weeks before the fair.

- Abstract – a free-form paragraph or two
- Quad Chart – a one-page structured summary of the project
- Project Presentation – a mini display for the project limited to 12 pages
- Public Video – prepared by the student for the general public
- ISEF forms

In-Person Resources

These resources are located at the project display itself in the Exhibit Hall, and will be available for judges to review after registering on Tuesday, the day before judging.

- Abstract – same as above
- Poster – the traditional project display
- Research notebook (optional)
- Research paper (optional)
- Demonstration Video (optional)
- ISEF forms

The Demonstration Video is distinct from the Public Video and is uncommon because it is used only to provide a demonstration where doing so in the Exhibit Hall is impractical.

Productive Preparation

Properly preparing for your interviews with the students takes time, but we do not want you to have to spend more time than necessary on any one project, nor to review more material than necessary, in order to gain a sufficiently complete understanding of each project. Students are encouraged to design all of their materials listed above in a clear and informative manner so

that you will be able to understand each project as quickly as possible, and thereby help to make your forthcoming discussions with the students more productive.

In this phase of preparation before speaking with the students, your job is:

- Gather a basic understanding of the project, and
- Formulate questions for the interview.

While it is natural during this process to form an impression about the quality of each project, and thereby develop a preliminary sense of how you will eventually rank the projects relative to each other, you should take care not to let that general sense become a prejudice coloring your views. It is not uncommon for the interviews to discover impressive work hiding behind rather modest early indications, and conversely, for impressive *Project Presentations* to distract from more pedestrian accomplishments by the students themselves.

First Approach

When you first consider any project, the abstract may appeal to you as the shortest description of the project. While it is short, it is also unstructured which means every student produces a unique insight into their work, and not always one that is productive for the judges. Certainly you should read the Abstract at some point, but you will likely find the structured documents of the Quad Chart and the *Project Presentation* more useful in your preparations.

You can do a quick initial triage of either of these documents by looking first at the hypothesis, engineering goals, or objectives, and then jump to the conclusions. Do they match? At this point, avoiding detail will enable a quick sense of the project's actual accomplishments, without the distractions of motivation or perceived future impact if the work is continued indefinitely.

Then skim the body of the project, avoiding details, but with the goal of understanding whether the work itself supports the conclusions. If these things do not match, consider why they do not. This superficial review quickly gives a good idea of the finalist's research without getting bogged down with issues not core to your evaluation.

The Deep Dive

Having gotten The Big Picture of a given project, it is then time to consider details. In doing so, remember these important features:

- **Good science:** Good science is not indicated by whether the project was done at home or in a lab, but by the merits of the project itself.
- **Creativity:** Is there something new, unique or different? Is the student clever? A creative project demonstrates imagination and inventiveness. Such projects often offer different perspectives that can reveal new thinking. Judges should place emphasis on research outcomes in evaluating creativity. At ISEF, projects should be original research or a creative take on something previously published in the literature.
- **Work:** The basis of science is reproducibility. Subjects, trials or sample size should be appropriate to the research. If possible, experiments should be run two or more times. If an experiment takes months to run, then it may only be done once. If trials can be run in a short period of time, making hundreds of trials is reasonable. One might test hundreds of growth plates, plants grown, etc. More is always better. The simple,

random sample is the standard for selecting a population to study. Calculating an appropriate size sample is simple. For large populations, ~370 subjects is the minimum sample size. There is a significant literature for calculating sample size for clinical trials. NIH literature suggests minimum sample size of 90, with 90 controls.

- **Their own work:** It is a challenge to determine what the student did and what was done by others. Don't be afraid to ask. For work done outside the student's home, be sure to review ISEF Form 1C.
- **Reproducibility:** The basis of science is reproducibility. If possible, experiments should be run 2 or more times. An experiment taking months to run can realistically only be done once. If a trial takes a short period of time, making a large number of trials is reasonable.
- **Statistical analysis:** All projects should have a reasonable statistical analysis, if appropriate. At minimum, there should be basic p or t values. Regression analysis is getting to be used more. The student should understand the concept of analysis of variance (ANOVA), and understand what they have done.
- **References:** References may be on the board or in the paper or lab notebook. They should be properly written. They should usually include some recent literature. Wiki is never used as a reference. They should be in a proper format, but any of the accepted formats is acceptable at ISEF.
- **Presentation:** Content is paramount, but good presentation may suggest an attention to detail that belies good content. While presentation is an important criterion in judging, do not let showy materials take the place of a good presentation and a meaningful interaction with the judge. Substance is critical. Writing should be clear and professional. Figures and tables should be clearly labeled with the appropriate units and axes indicated. Flow of the information should be clear leading up to how the student developed their conclusion or produced their engineering apparatus.

Research papers are optional for ISEF, though they are required by some country/state/regional/local fairs. The research paper should be in a proper scientific format. It should look like it is ready to be submitted for publication. Some of these finalist's projects will already be published. Papers that do not meet this standard cast doubt on the quality of the research. Check the references. They should be in a proper academic format. There are several standard formats, but any are acceptable for ISEF. Wiki is never an appropriate reference. Poor references may suggest poor work.

The notebook, abstract, research paper, poster, and other materials should all be consistent.

Point of Emphasis

The difference between a winning project and just another project is attention to detail. The review without the student present should prepare the judge for the interview, help raise questions to be asked at interview, and give an early sense of the level of quality of each project.